

The importance of pH in human biology

From protonation states to blood buffering

Case report

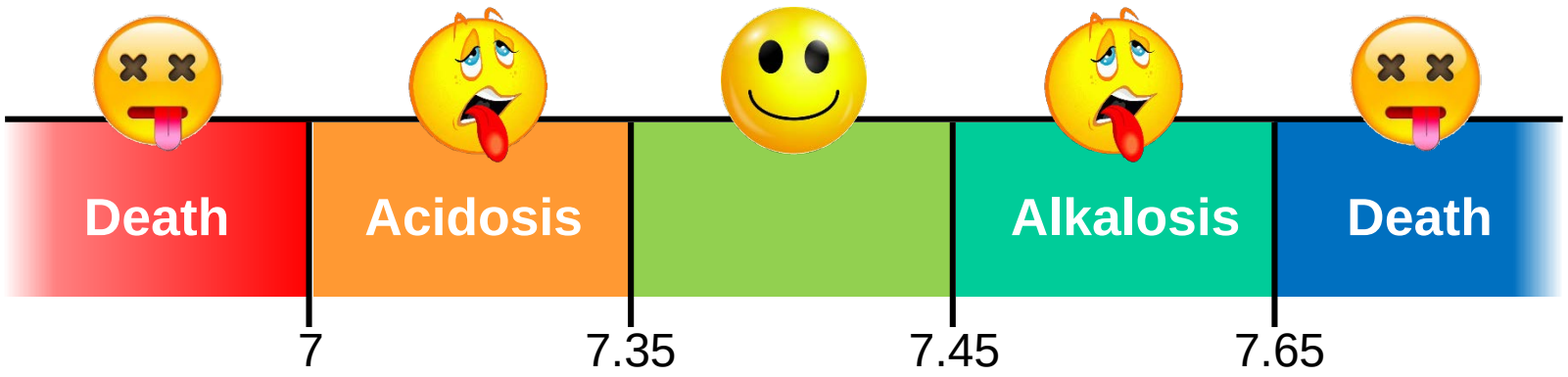
A 45-year-old man was admitted with nausea, vomiting, and abdominal pain, with a pCO_2 of 85.5 mmHg, pO_2 of 43.5 mmHg, pH of 7.61, and plasma bicarbonate concentration of 82 mM.

High pCO_2

Low pO_2

High pH

High bicarbonate



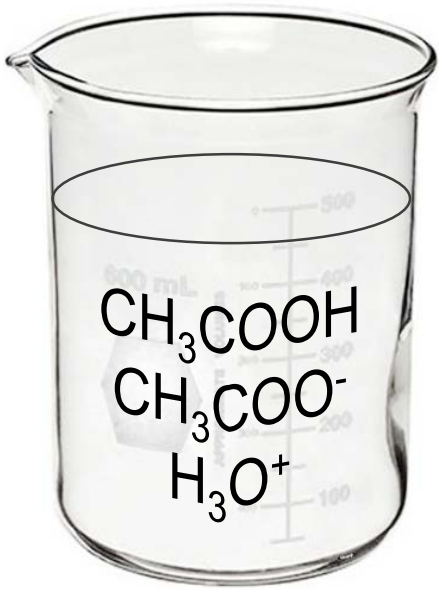
Javaheri S and Nardell EA. *Br Med J* 1981



Why is pH physiologically relevant?

Why does a very small change in blood pH have such a strong impact on human health?

Henderson-Hasselbalch equation



Acetic acid in water

$$CH_3{-}COOH + H_2O \rightleftharpoons CH_3{-}COO^- + H_3O^+$$

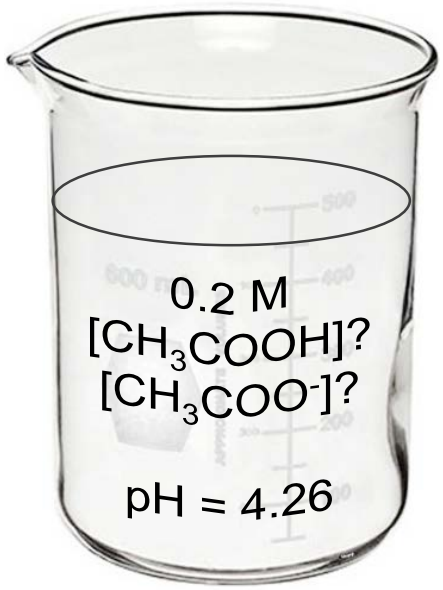
Acetic acid — Acetate

$$K_a = \frac{[Ac^-]\,[H^+]}{[AcH]}$$

Conjugate pair!

$$pH = pK_a + \log \frac{[Ac^-]}{[AcH]}$$

$$pH = pK_a + \log \frac{[Base]}{[Acid]}$$



$$CH_3{-}COOH + H_2O \rightleftharpoons CH_3{-}COO^- + H_3O^+$$

Acetic acid — Acetate

$$pH = pK_a + \log\frac{[Base]}{[Acid]}$$

$4.26 = 4.74 + \log [Ac^-]/[AcH]$

$[Ac^-]/[AcH] = 0.33$

$[Ac^-]/(0.2-[Ac^-]) = 0.33$

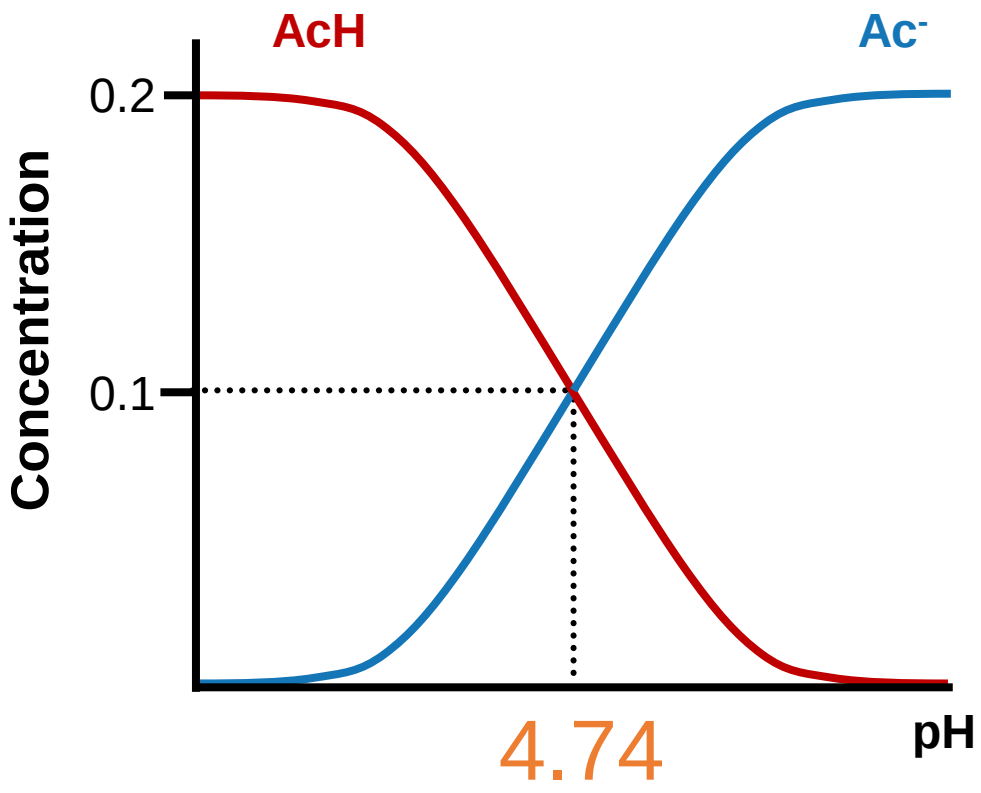
$[Ac^-] = 0.05$ M $[AcH] = 0.15$ M

$$CH_3{-}COOH + H_2O \rightleftharpoons CH_3{-}COO^- + H_3O^+$$

Acetic acid — Acetate

$$pH = pK_a + \log\frac{[Base]}{[Acid]}$$

pH	[Acetic acid] (M)	[Acetate] (M)
1	0.19996	0.00004
1.8	0.1997	0.0002
3	0.1964	0.0036
3.76	0.181	0.019
4.26	0.15	0.05
5.2	0.051	0.149
6	0.013	0.187
7	0.001	0.199
8.3	0.00006	0.199994



$CH_3-COOH + H_2O \rightleftharpoons CH_3-COO^- + H_3O^+$

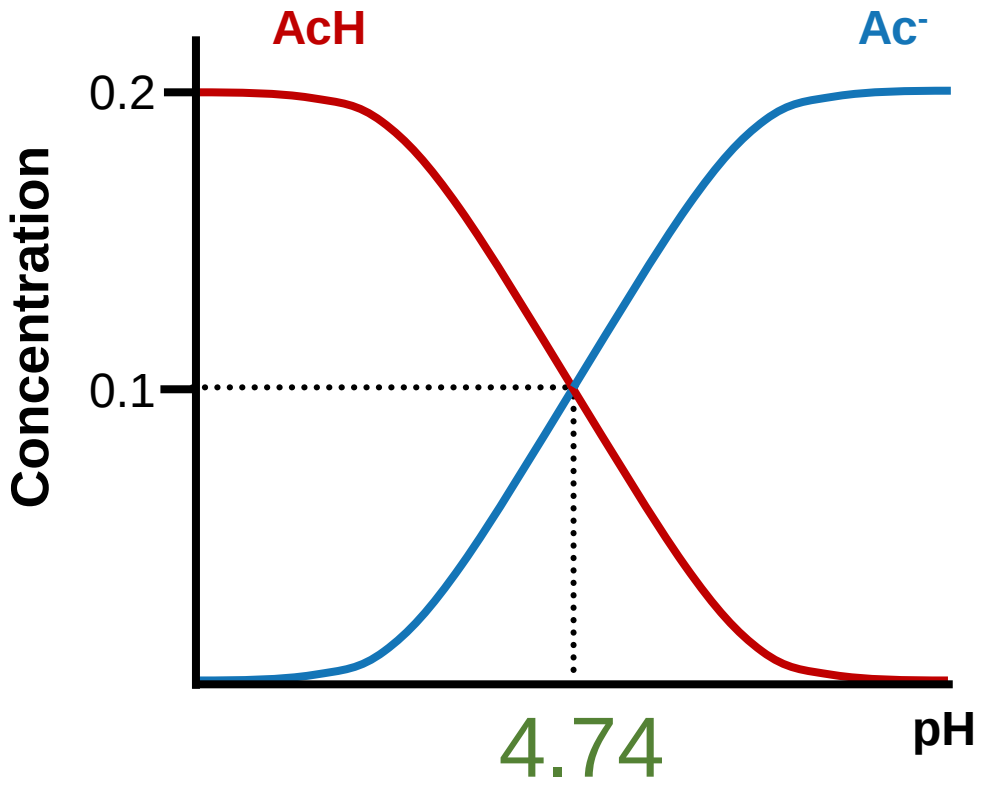
Acetic acid — Acetate

$$pH = pK_a + \log \frac{[Base]}{[Acid]}$$

$$pH < pK_a \rightarrow \frac{[Ac^-]}{[AcH]} < 1 \rightarrow [Ac^-] < [AcH]$$

$$pH = pK_a \rightarrow \frac{[Ac^-]}{[AcH]} = 1 \rightarrow [Ac^-] = [AcH]$$

$$pH > pK_a \rightarrow \frac{[Ac^-]}{[AcH]} > 1 \rightarrow [Ac^-] > [AcH]$$



Acids and bases in organic biomolecules

Carboxyl group (-COOH)

Sulfhydryl (thiol) group (-SH)

Phosphoryl group ($-PO_4H_2$)

Hydroxyl group (-OH)

Acids

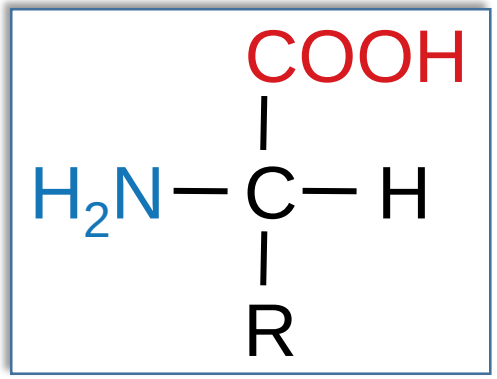
$-AH + H_2O \leftrightarrows -A^- + H_3O^+$

Amino group ($-NH_2$)

Base

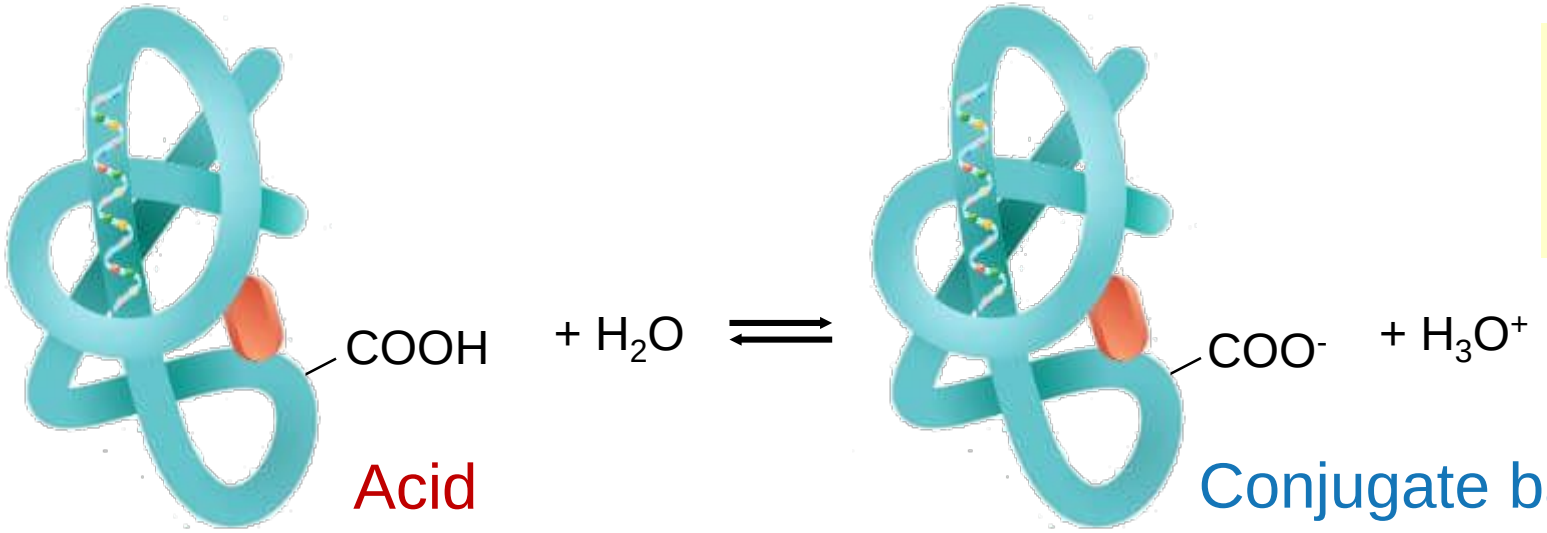
→ Conjugate acid: $-NH_3^+$

$-NH_3^+ + H_2O \leftrightarrows -NH_2 + H_3O^+$



Amino acid	Abbreviation/ symbol	M_r	pK_a values: pK_1 (—COOH)	pK_2 ($—NH_3^+$)	pK_R (R group)	pI	Hydropathy index*	Occurrence in proteins (%)†
Nonpolar, aliphatic R groups								
Glycine	Gly G	75	2.34	9.60		5.97	−0.4	7.2
Alanine	Ala A	89	2.34	9.69		6.01	1.8	7.8
Proline	Pro P	115	1.99	10.96		6.48	1.6	5.2
Valine	Val V	117	2.32	9.62		5.97	4.2	6.6
Leucine	Leu L	131	2.36	9.60		5.98	3.8	9.1
Isoleucine	Ile I	131	2.36	9.68		6.02	4.5	5.3
Methionine	Met M	149	2.28	9.21		5.74	1.9	2.3
Aromatic R groups								
Phenylalanine	Phe F	165	1.83	9.13		5.48	2.8	3.9
Tyrosine	Tyr Y	181	2.20	9.11	10.07	5.66	−1.3	3.2
Tryptophan	Trp W	204	2.38	9.39		5.89	−0.9	1.4
Polar, uncharged R groups								
Serine	Ser S	105	2.21	9.15		5.68	−0.8	6.8
Threonine	Thr T	119	2.11	9.62		5.87	−0.7	5.9
Cysteine	Cys C	121	1.96	10.28	8.18	5.07	2.5	1.9
Asparagine	Asn N	132	2.02	8.80		5.41	−3.5	4.3
Glutamine	Gln Q	146	2.17	9.13		5.65	−3.5	4.2
Positively charged R groups								
Lysine	Lys K	146	2.18	8.95	10.53	9.74	−3.9	5.9
Histidine	His H	155	1.82	9.17	6.00	7.59	−3.2	2.3
Arginine	Arg R	174	2.17	9.04	12.48	10.76	−4.5	5.1
Negatively charged R groups								
Aspartate	Asp D	133	1.88	9.60	3.65	2.77	−3.5	5.3
Glutamate	Glu E	147	2.19	9.67	4.25	3.22	−3.5	6.3

Lehninger 4th ed.



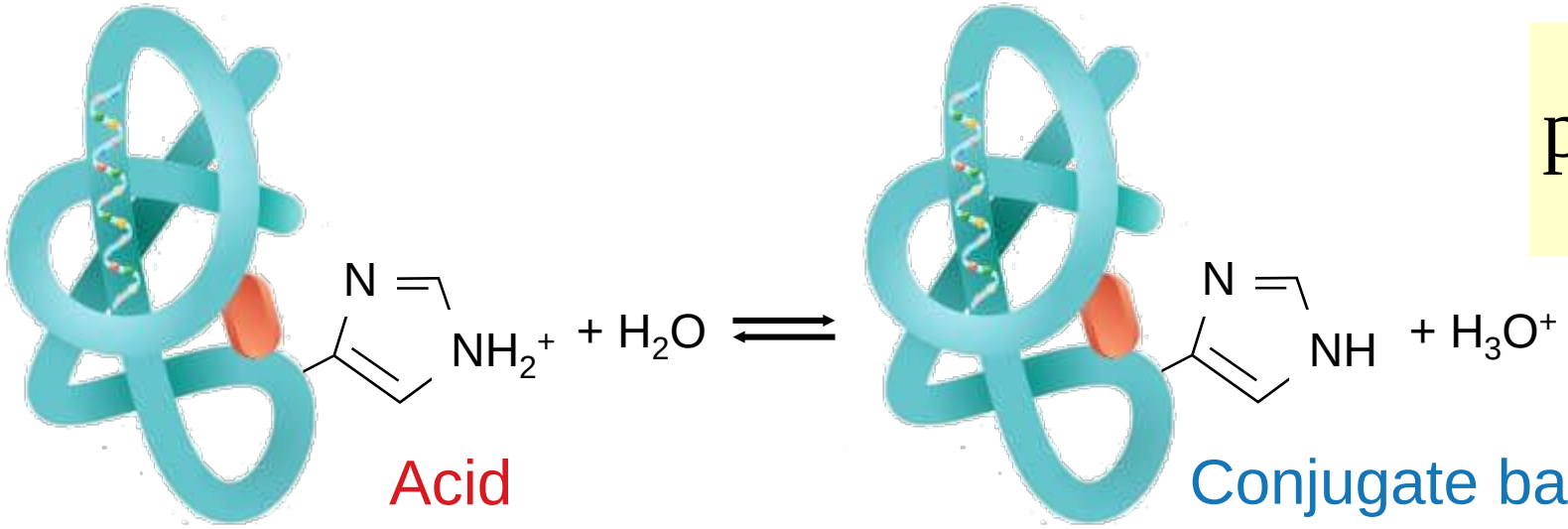
$$pH = pK_a + \log \frac{[Base]}{[Acid]}$$

What proportion of the -COOH group is deprotonated under physiological conditions (pH 7.4)? (pK_a = 3.6)

7.4 = 3.6 + log [-COO^-]/[-COOH]

[-COO^-]/[-COOH] = 6310

99.9%



$$pH = pK_a + \log \frac{[Base]}{[Acid]}$$

What proportion of the imidazole group is deprotonated? (pK_a = 7.2)

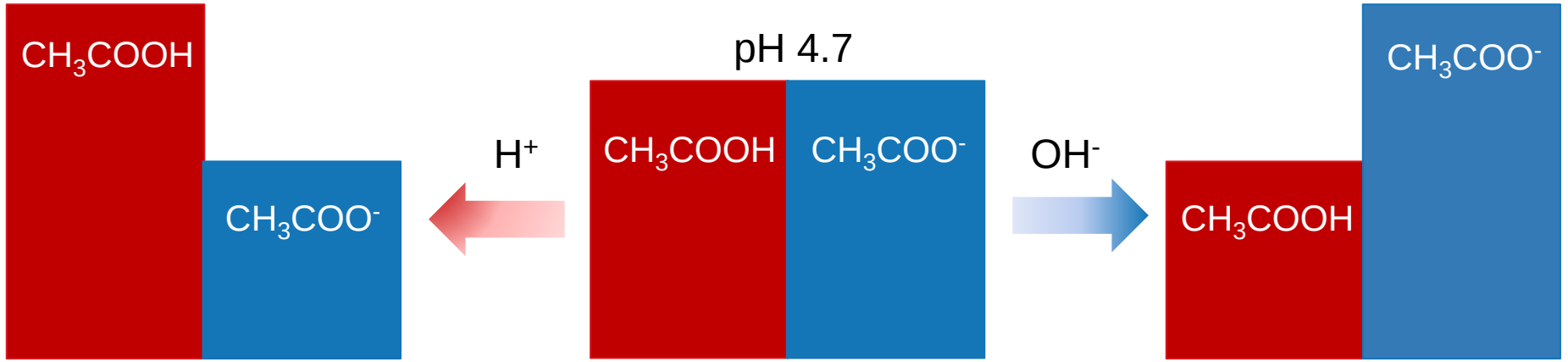
pH	[Base] / [Acid]	% Base	% Acid
7.4	1.6	61	39
7.6	2.5	72	28
8	6.3	86	14
9	63	98	2

How does my body protect me from dangerous changes in blood pH?



Buffer solution

$AcH + H_2O \longleftrightarrow Ac^- + H_3O^+$



If an acid is added

If a base is added

A solution with a mixture of a weak acid and its conjugate base is able to neutralize the addition of acids and bases, keeping the pH stable

Blood is a buffer solution

- To maintain the pH of blood in a normal range, our body uses different buffer systems.
- The main buffer system in our blood is the carbonic acid/bicarbonate buffer.

$$CO_2 + H_2O$$

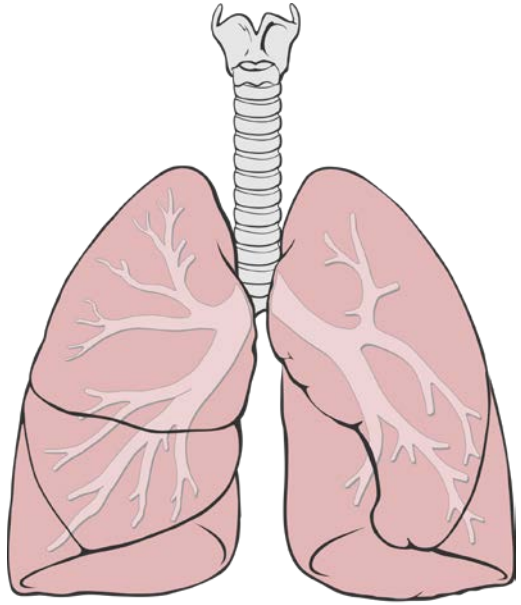
Carbonic anhydrase ↕

$$H_2CO_3 + H_2O \leftrightarrow HCO_3^- + H_3O^+ \quad pK_a = 6.1$$

Can I voluntarily modify my blood pH?



Respiratory control of pH



$$CO_2 + H_2O$$

$$\updownarrow$$

$$H_2CO_3 + H_2O \leftrightarrow HCO_3^- + H_3O^+$$

Hypoventilation reduces blood pH

Hyperventilation increases blood pH

Case report

A 45-year-old man was admitted with nausea, vomiting, and abdominal pain, with a pCO_2 of 85.5 mmHg, pO_2 of 43.5 mmHg, pH of 7.61, and plasma bicarbonate concentration of 82 mM.

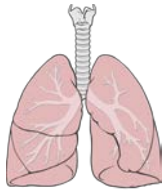
High pCO_2

Low pO_2

High pH

High bicarbonate

$$CO_2 + H_2O$$



$$\updownarrow$$

$$H_2CO_3 + H_2O \leftrightarrow HCO_3^- + H_3O^+$$



The patient was hypoventilating to compensate his metabolic alkalosis!

Javaheri S and Nardell EA. *Br Med J* 1981

Bonus question

- The total concentration of carbonic acid/bicarbonate in blood is approximately 45 mM. Calculate the final pH if 1 ml of HCl 1 M is added to a liter of a solution containing 45 mM of this buffer at pH 7.4, knowing that:

$$CO_2 + H_2O \leftrightarrow HCO_3^- + H_3O^+ \qquad pK_a = 6.1$$

- What would be the pH if the same amount of acid was added to pure water?